

Checkpoint

12.5 Part 2: Planes

Problem 1. *What two things do you need for a plane in 3D?*

Problem 2. *Find the normal to the following plane.*

$$x + 6y - 4z = 12$$

Problem 3. *If I gave you two (non-parallel) lines in a plane, how would you find the equation of the plane?*

Problem 4. *Find the equation of the plane with normal $2\hat{i} - 5\hat{j} + \hat{k}$ which goes through the point $(5, -3, 0)$.*

Problem 5. *Are the following planes perpendicular?*

$$2(x - 100) + 2(y - 200) + 2(z - 300) = 0$$

and

$$(x + 100) + (y + 200) + (z + 300) = 0$$

1 Answers

1. A point and a normal
2. $\langle 1, 6, -4 \rangle$
3. Use the point on either line for the **point** and take the cross product of the directions of the lines for the **normal**.
4. $2(x - 5) - 5(y + 3) + z = 0$
5. No $\langle 2, 2, 2 \rangle \cdot \langle 1, 1, 1 \rangle \neq 0$